

HyTouch: A Bio-inspired Hybrid Multimodal Tactile Sensor with High Spatial and High Temporal Resolution

Abstract—Robotic tactile sensing has been fundamentally constrained by a “tactile trilemma”: a trade-off among high-resolution spatial sensing, high-frequency dynamic response, and spatially distributed force sensing. This paper introduces HyTouch, a novel sensor with a generalizable hardware integration approach that addresses this trilemma. The proposed solution physically collocates a high-frequency array layer into a high-resolution visual-tactile sensor which provides both dynamic and spatially distributed force data. This co-location is made possible by the careful design of an optical system with real-time signal processing technique that resolves inter-modality interference. A prototype sensor made with the proposed approach is validated in two representative robotic tasks. Firstly, in a high-speed cable-following task, a simple controller exploits the sensor’s high-frequency force distribution to achieve rapid (300 Hz) positional tracking - a challenge for sensors with single-channel force feedback. With the proposed sensor prototype, a maximum speed of 24 cm/s for the cable following task is attained with a significant reduction in corrective adjustments. Secondly, in a tool-mediated material classification task, a simple sensor-fusion model is benchmarked against models using data from respective individual modalities. The fused model achieves 99.25% accuracy, outperforming individual visual-tactile (80.00%) and force-tactile (91.13%) modalities, which demonstrates the information complementarity between the two. The HyTouch prototype demonstrates that the integrated approach provides a hardware solution to a fundamental trade-off, enabling richer, more robust perception for complex robotic interaction.

Index Terms—Tactile Sensor, Multimodal Fusion, Vision-Based Sensors, Force Sensing, Bio-inspired Robotics, Robotic Manipulation.

I. INTRODUCTION

TACTILE sensing is essential for dexterous robotic manipulation and supports a wide range of applications, including stable grasping, contact-rich manipulation, state estimation of deformable objects, and tool-mediated interaction [1]–[3]. To endow robots with the sense of touch, a wide variety of tactile sensors have been proposed, including vision-based designs (e.g. GelSight [4], TacTip [5], DIGIT [6]) and electronic designs (e.g. BioTac [7], XELA uSkin [8], GTac [9]). Although these sensors differ in their transduction principles, deployment scenarios, and performance characteristics, they still often face the fundamental “tactile trilemma”, that is, achieving high-resolution spatial sensing, high-frequency dynamic response, and spatially distributed force sensing in a single device remains inherently challenging [10], [11]. Specifically, vision-based tactile sensors (VBTS) capture fine textures, but their effective information rate is typically low (around 20 to 30 Hz) due to the needs of image processing [12]. Electronic tactile sensors (ETS) offer high-frequency responses for vibration and slip detection and can provide spatially distributed force information through array structures for contact-state regulation. However, their spatial resolution

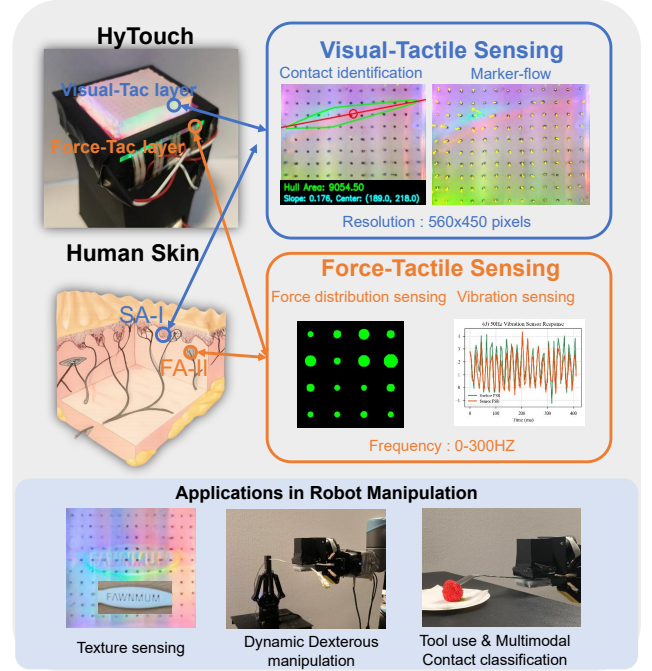


Fig. 1: The core concept of our multimodal integration method, which combines high-resolution visual-tactile data with high-frequency force-tactile feedback to enable robust perception in dynamic tasks.

remains limited, preventing them from directly capturing fine contact geometry or surface texture [13]. By contrast, human skin co-locates multiple receptor types within the same region, enabling it to achieve all three capabilities simultaneously and thereby overcome the tactile trilemma [14], [15].

Importantly, the trilemma concerns performance over the number of modalities. In practice, array-based electronic sensors (resistive/capacitive/magnetic) can simultaneously provide high-frequency response and spatially distributed force distribution maps, making them well suited for reactive control, slip detection, and vibration sensing [2], [16]. However, they typically lack the ability to perceive fine surface details. In contrast, vision-based sensors can provide high-resolution surface information, but suffer from lower effective temporal response due to sensing and processing latency. As such, the objective of this work is to propose a novel solution that integrates high spatial resolution visual tactile sensing and high temporal resolution force distribution sensing into a single tactile sensor, thereby addressing all three axes of the tactile trilemma.

To this end, a generalizable hardware integration approach

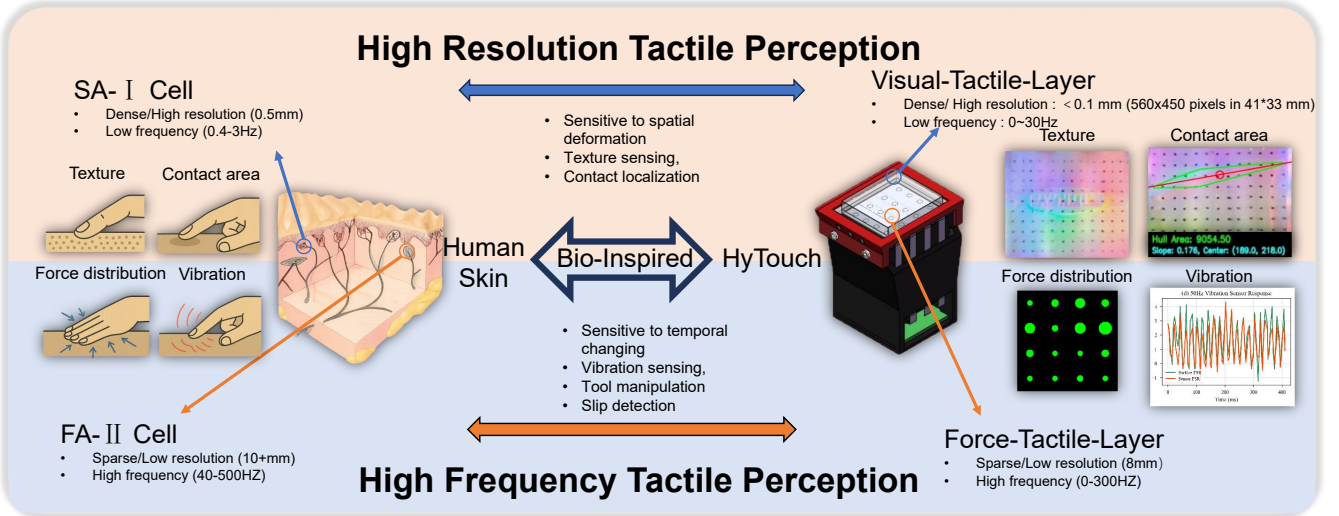


Fig. 2: Illustration of the multimodal integration concept, inspired by the complementary roles of SA-I and FA-II mechanoreceptors in human skin. Our method combines a vision-based layer (inspired by SA-I receptors) for High-Resolution Spatial Sensing with a sparse, high-frequency force-sensing array (inspired by FA-II receptors) for High-Frequency Dynamic Response and Spatially Distributed Force Sensing.

is proposed that explicitly combines the three tactile capabilities through co-location and decoupling. Specifically, a high-frequency force sensing (e.g., resistive) array and a visual-tactile gel layer are stacked within the same sensing region to jointly cover all three capability dimensions. At the same time, the optical and signal-processing pipelines are carefully developed to minimize mutual interference. A compact two-mirror optical path together with a side-entering light-guiding lens is designed to prevent the force-sensing array from occluding or corrupting the visual pathway. In addition, an in-sensor, gradient-based image stitching pipeline, including undistortion, homography warping, and Poisson blending, is proposed to eliminate the visible seams in the reconstructed tactile image. As a result, the proposed sensor design simultaneously delivers high-resolution surface perception alongside high-frequency, spatially distributed force feedback.

With the proposed principle, we design HyTouch, a co-located and decoupled bio-inspired multimodal tactile sensor that integrates a vision-tactile sensing layer with a force-sensing array. The sensor incorporates a compact reflective mirror system, a light-guiding lens for uniform side illumination, and a reconfigurable housing that enables precise alignment. In addition, an optical-path simulation and design tool is developed and employed to validate blind-spot elimination and to guide the selection of key mechanical design parameters.

Quantitative evaluations are conducted to assess visual-tactile reconstruction quality, calibrate the force-tactile sensing, and characterize elastomer-induced attenuation of vibration signals transmitted from the visual-tactile layer to the force sensing array. The HyTouch is then benchmarked on two representative tasks that require multimodal tactile sensing: (i) high-speed cable following, where a simple controller leverages high-frequency, spatially distributed force feedback for reactive tracking while using high-resolution

visual-tactile information for grip-pose adjustment; and (ii) tool-mediated material classification, in which a lightweight fusion model demonstrates superior performance compared to single-modality baselines.

The main contributions of this paper are therefore:

- A bio-inspired generalizable co-located and decoupled visual-force-tactile approach that addresses all three tactile capability axes: high-resolution surface imaging, high-frequency dynamic response, and spatially distributed force sensing. This architecture comprises (i) a compact two-mirror optical system for unobstructed multi-view imaging, (ii) a side-entering light-guiding lens that suppresses stray reflections and avoids array-induced glare, and (iii) an in-sensor, gradient-domain fusion pipeline for seamless occlusion removal.
- The design and implementation of the **HyTouch** sensor, together with a light-path simulation and design tool for HyTouch-like sensors. This tool explicitly models visual-tactile occlusions caused by the force-tactile layer and supports the joint optimization of mirror geometry and force-tactile array configuration.
- Quantitative evaluation of cross-modality decoupling, including optical-path simulation and visual reconstruction metrics, vibration transmission analysis through the elastomer layer, and force-response calibration of the force-tactile array.
- Experimental demonstrations on two representative robotic tasks: (i) high-speed cable following and (ii) tool-mediated material classification, showing that the proposed sensor design and multimodal tactile fusion outperform single-modality sensing and address the tactile trilemma.

The rest of this paper is organized as follows. Section II reviews related work. Section III details the design of the HyTouch sensor. Sections IV and V present the sensor

prototype and characterization, as well as the robotic use cases, respectively. Finally, Section VI concludes the paper.

II. RELATED WORK

A. The Role of Tactile Sensing in Robotics

In many robotic manipulation tasks, vision alone becomes unreliable due to occlusions or lighting variations, while the force feedback from the robot end-effector often lacks the spatial resolution needed to accurately infer the local contact states [17]. To achieve human-like dexterity, tactile sensing has emerged as a key frontier in robotics research [10], [17].

Tactile perception empowers robots by providing rich-contact information across multiple dimensions. Firstly, it enables the estimation of intrinsic object properties, including texture [18], compliance [19], and even liquid properties [20]. Secondly, it enables precise pose and shape estimation [21], [22], which is especially important for manipulating deformable objects such as cables, where visual state estimation remains challenging [1]. Thirdly, tactile feedback supports complex physical interactions, from detecting incipient slip for grasp stabilization [2] to enabling dexterous in-gripper manipulation through extrinsic contact sensing [22], [23]. This capability further extends to robust control [24] and to non-prehensile manipulation tasks where exploiting environmental constraints is essential [3], [25], [26]. Recent studies have further highlighted the critical role of high-dynamic tactile feedback [27].

B. The Tactile Trilemma

Most of the applications discussed above demonstrate that dexterous manipulation is not enabled by any single sensing modality, but rather by a collection of complementary capabilities as detailed below.

- **High-Resolution Surface Sensing:** Essential for perceiving object properties (e.g., texture) and estimating the continuous state of deformable objects (e.g., cables).
- **High-Frequency Dynamic Response:** Critical for capturing transient events, such as micro-vibrations during incipient slip or rapid dynamics.
- **Spatially Distributed Force Sensing:** Required for resolving complex contact force/torque (F/T) states and estimating extrinsic states during tool-mediated or environment-aware manipulation.

Ideally, a robotic tactile sensor should integrate all three capabilities simultaneously to resolve the aforementioned trilemma. A natural approach is to mimic the sensory fusion of human skin by co-locating hybrid modalities within a unified sensing structure. However, achieving such integration remains difficult due to fundamental hardware constraints, particularly the challenge of physically co-locating distinct sensing elements without inducing mutual interference or occlusion. Consequently, existing sensor designs typically satisfy only one or two of these capabilities, leading to the trade-offs in the “Tactile Trilemma”. To better understand this gap, the following section reviews the evolution of tactile sensing from a chronological perspective, analyzing how different sensor

generations have addressed specific aspects of this trilemma while compromising on others.

Electronic Arrays. The foundational phase of tactile sensor development focused on establishing dynamic response and force distribution using electronic transduction. This type of tactile sensors includes force-sensing arrays such as NeuTouch [16], capacitive tactile skins widely used as robot skin on platforms like iCub [13], [28], and magnetic sensors like uSkin [8] and ReSkin [29]. Advanced versions, such as NUskin, can achieve sampling rates as high as 4 kHz [30], while the magnetic sensors offer high linearity in shear force detection [8].

These sensors successfully satisfy *High-Frequency Dynamic Response* and *Spatially Distributed Force Sensing*. However, they fundamentally fail *High-Resolution Surface Sensing*. As noted in [10], these arrays typically provide low data density compared to optical methods. A sparse array (e.g., taxels spaced millimeters apart) is physically unable to directly resolve the fine surface textures required for detailed geometry perception.

Vision-Based Tactile Sensors (VBTSs). Driven by the miniaturization of cameras and cost reductions, recent years have witnessed an explosion in VBTS [17], [31]. This approach prioritizes *High-Resolution Surface Sensing* by utilizing an internal camera to capture the deformation of a soft elastomer skin [4].

This field is dominated by two primary optical principles. The first category relies on photometric stereo or shading cues to reconstruct fine surface geometry. The canonical example is GelSight [4], which achieves extremely high spatial resolution. This design has inspired a rich family of variants tailored for specific robotic morphologies, including the compact DIGIT [6] for in-hand manipulation, the finger-shaped GelTip [32], and the slim-profile GelSlim [33] for gripper integration. Novel designs such as TwinTac [34] and DTactive [35] have also been introduced to improve sensing range and active perception capabilities. Other innovations include HaptiTemp [36] for thermal sensing and ThinTact [37] for ultra-thin form factors. Specialized designs such as GelFinger [38] and TouchRoller [39] have also been proposed for multi-angle and large-area assessment. The second category employs structured-marker-based tracking, where the camera tracks the displacement of internal pins to infer contact. Notable examples include the biomimetic TacTip family [5], [40]. Moreover, recent advancements like ViTacTip [41] attempt to fuse vision and tactile modalities more closely.

These tactile sensors set the benchmark for spatial resolution, capable of discerning micro-textures that electronic arrays cannot detect. However, they compromise on *High-Frequency Dynamic Response*. As highlighted in [17], the computational overhead of inferring force from images is significant. While raw camera frame rates can be high, the heavy post-processing required to extract meaningful physical data (force, shear, slip) constrains the *effective information rate* to less than 100 Hz and even to below 20 Hz in many cases [12]. The latency of VBTS makes them unsuitable for the high-speed feedback loops required in rapid dynamic manipulation. This limitation motivates the hybrid approach proposed in this paper.

Early Multimodal Exploration. In the 2010s, sensors like the SynTouch BioTac attempted to bridge gaps by integrating multiple modalities. It measures impedance changes in a conductive fluid for deformation sensing, alongside hydrophones for vibration and thermistors for temperature [42]. This integration allowed for the detection of micro-vibrations and texture classification via temporal signals [43].

While the BioTac satisfies *High-Frequency Dynamic Response* (via hydrophones), it effectively acts as a complex “point sensor” regarding spatial distribution. It fails both *High-Resolution Surface Sensing* (lacking pixel-level geometry) and true *Spatially Distributed Force Sensing* (inferring contact location rather than measuring a distributed pressure map). Furthermore, the high cost and complexity of fabrication have limited its widespread adoption. [6].

Hybrid Modality Integration. Recent outlooks indicate that hybrid approaches are emerging to address the limitations of single-modality approaches, such as DIGIT 360 [44]. A common strategy is to augment a visual core with a single-channel force or pressure element [44], [45]. These hybrid designs successfully combine *High-Resolution Surface Sensing* (vision) with *High-Frequency Dynamic Response* (single-channel pressure). However, they critically fail *Spatially Distributed Force Sensing*. A single pressure reading implies a “point sensor” configuration regarding force which cannot distinguish between a concentrated point force and a distributed load, nor can it track the location of a dynamic contact at high frequency. EveTac [46] explores neuromorphic event-based cameras to improve temporal resolution, but this improvement sacrifices the global surface sensing and static sensing ability of VBTS.

One recent work presented in [47] attempts to combine magnet arrays with VBTS by laying a magnet array below the deformable gel to improve the force measurement. However, the center of the array cannot be sensed by the Hall sensor due to the placement of the camera, and the magnets also cause occlusion between the deformation gel and the camera, which sacrifices the marker flow function of VBTS.

In summary, the tactile trilemma remains unsolved in existing tactile sensors, despite the growing need for richer, more capable, and more integrated tactile sensing.

III. SENSOR DESIGN

To resolve the tactile trilemma at the hardware level, a novel tactile sensor design leveraging on physical *co-location* and optical *decoupling* is proposed. The designed sensor is named HyTouch, which synergistically fuses a high-resolution visual-tactile layer with a high-frequency force-sensing array within the same sensing volume. Unlike prior hybrid designs that simply place sensors side-by-side, HyTouch achieves true co-location by employing a custom-engineered optical system to resolve the inherent occlusion and interference between the vision and force-sensing layers. As illustrated in the exploded view in Fig. 3, this architecture allows the sensor to simultaneously capture fine surface textures and dynamic force distributions from the same contact interface, effectively bridging the gap between static detail and dynamic response.

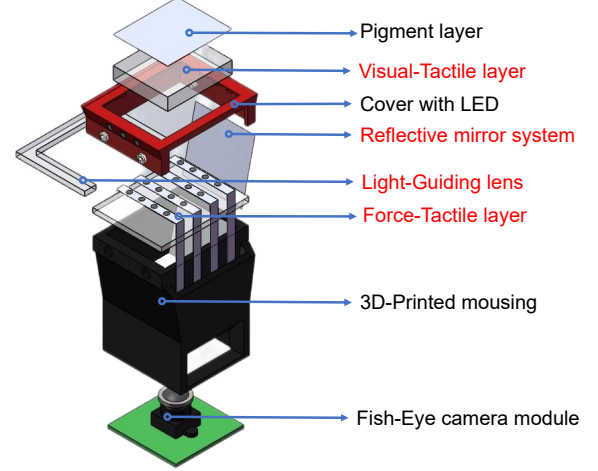


Fig. 3: Exploded view of the HyTouch sensor, illustrating the layered composition of the visual-tactile system, the force-sensing array, the multi-view optical system, and the housing.

The sensor configuration consists of the following key components:

- **Visual-Tactile Layer:** An elastomer layer with a pigmented top surface.
- **Force-Tactile Layer:** A array of force-sensing elements.
- **Mirror System:** A two-mirror system to provide multiple virtual camera views, resolving occlusion.
- **Light-Guiding Lens:** A structure to ensure uniform illumination and aid in component alignment.
- **RGB LED Set:** The lighting source to illuminate the gel layer to enhance texture visibility.
- **Camera Module:** A fisheye camera to capture the deformation of the gel layer.
- **3D-Printed Housing:** A modhousing that provides structural stability and precise alignment.

A. Bio-Inspired Sensor Architecture

Human skin senses touch using several types of receptors that sit in the same area, with each type doing a different job, some detect fine shapes and textures, while others respond quickly to vibrations and sudden changes [14], [15]. Inspired by this idea, HyTouch is designed with two functional layers that follow the same “division of labor”:

- **Visual-Tactile Layer (inspired by SA-I):** This layer emulates the function of Slowly Adapting Type I (SA-I) mechanoreceptors (Merkel discs). In human skin, SA-I afferents are characterized by high spatial density (approximately 0.5 mm spacing in fingertips) but limited temporal bandwidth (0.4 to 3 Hz), making them responsible for static form perception and texture resolution [48]. Similarly, the visual-tactile layer is designed by utilizing a soft elastomer (Smooth-On Solaris) with a pigmented surface to achieve *High-Resolution Surface Sensing*. By

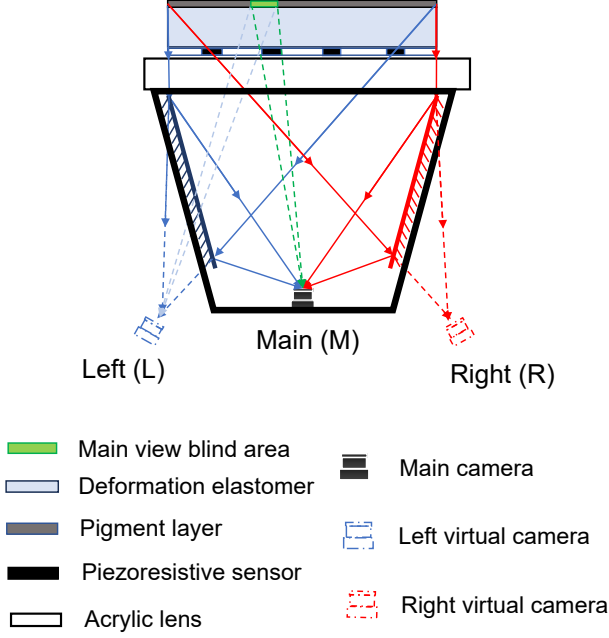


Fig. 4: Sensor optical design

using a camera system (like GelSight), fine spatial deformations can be captured with fidelity (e.g., <0.1 mm with 560×450 pixels) exceeding that of human touch. However, similar to the SA-I channel, its temporal response is constrained by the camera frame rate [about 30 fps (i.e., Hz)].

- **Force-Tactile Layer (inspired by FA-II):** This layer mimics the role of Fast Adapting Type II (FA-II) mechanoreceptors (Pacinian corpuscles). In human skin, FA-II afferents are sparsely distributed (receptive fields >10 mm) but possess high-frequency sensitivity (40 to 500 Hz), enabling the detection of vibrations, acceleration, and transient slip events [14].

Correspondingly, the force-tactile layer is designed as a sparse array of force-sensing elements. Its wider taxel spacing reflects the broader receptive fields of FA-II units, emphasizing coverage rather than fine spatial detail. Importantly, this layer delivers high-frequency dynamic response and spatially distributed force sensing, allowing it to capture rapid dynamic events that the visual-tactile layer cannot.

By physically co-locating the artificial “SA-I” and “FA-II” layers, the HyTouch achieves a synergistic perception capability that addresses the tactile trilemma like human skin. However, the force-tactile layer create occlusion to the visual-tactile layer due to this co-locating design. To address this issue, an innovative design of the visual-tactile subsystem is introduced and detailed in the following subsection.

B. Visual-Tactile Subsystem

The visual-tactile subsystem is responsible for capturing high-resolution surface information. Its design involves a

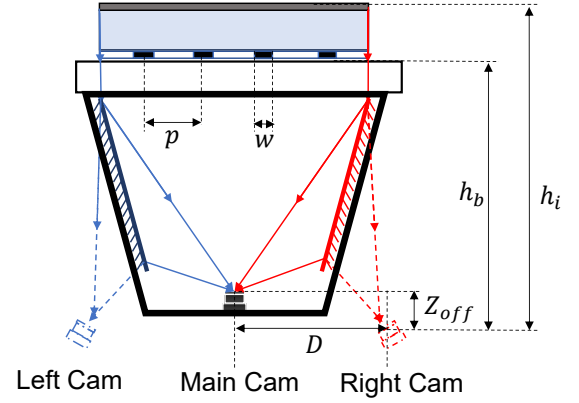


Fig. 5: Design parameters of the compact reflective mirror system.

tightly integrated set of mechanical, optical, and algorithmic components to overcome the physical challenges of a multi-modal design.

1) *Integrated Mechanical and Optical Design:* All the sensor’s components are housed within a 3D-printed structure designed for both stability and serviceability. A key feature is its modular design as shown in Fig. 3, which allows for easy disassembly and reassembly, ensuring consistent alignment of all optical and sensing components when a part, such as the elastomer, is replaced.

A challenge in integrating the two sensing layers is the physical occlusion caused by the non-transparent force-sensing array, which obstructs the camera’s view of the elastomer above it. To overcome this, a novel and compact multi-view optical system is designed.

2) *Reflective Mirror System and Simulation-based Design Tool:* To provide multiple views of the visual-tactile layer to a single camera, two angled reflective mirrors are mounted on the housing. This mirror configuration creates three virtual viewpoints arranged spatially within the captured frame: one on the left, one in the center, and one on the right. By fusing these views into a unified image using the reconstruction pipeline detailed in Section III-B4, the occlusions originating from the force-tactile layer can be eliminated. The key design parameters are summarized in Fig. 5.

a) *Geometric Constraints for Mirror Placement:* Fundamentally, the design of the mirror system is a *coverage search problem* to ensure zero blind spots. Let the imaging plane be denoted as $\mathcal{S}$ and the set of opaque force-sensing elements on the intermediate blocking layer as $\mathcal{O}$. Given a main camera $\mathbf{c}_m$, the objective is to determine the optimal configuration of virtual cameras $\mathbf{c}_v$ (generated by mirrors) such that every point $\mathbf{p} \in \mathcal{S}$ is visible to at least one camera. Mathematically, let $L(\mathbf{c}, \mathbf{p})$ be the line segment connecting a camera $\mathbf{c}$ and a target point $\mathbf{p}$. The visibility constraint requires that the intersection of this line of sight with the obstacle set is empty:

$$\forall \mathbf{p} \in \mathcal{S}, \exists \mathbf{c} \in \{\mathbf{c}_m\} \cup \mathcal{C}_v \text{ s.t. } L(\mathbf{c}, \mathbf{p}) \cap \mathcal{O} = \emptyset. \quad (1)$$

In the general case where $\mathcal{O}$ comprises arbitrarily distributed elements, satisfying (1) becomes a complex search prob-

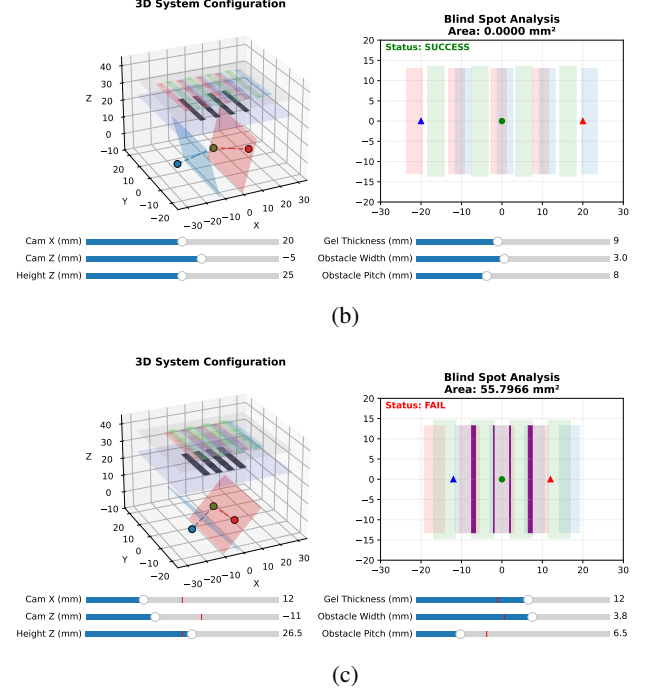
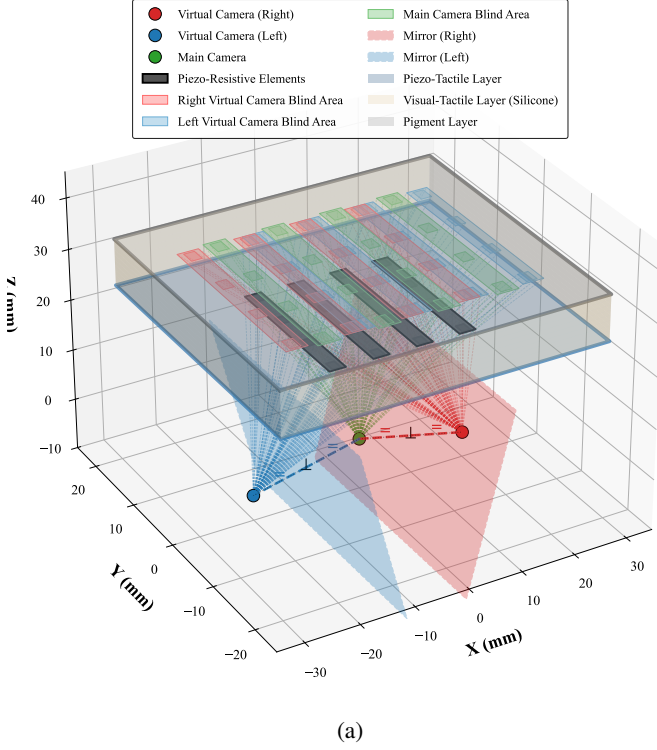


Fig. 6: Validation using the interactive design tool: (a) visualization of sensor elements within the simulator; (b) a successful configuration showing full coverage; (c) a failed configuration where improper parameters lead to blind spots (highlighted in purple).

lem without a closed-form solution. Critically, our proposed simulation-based design tool is capable of solving this general problem by performing iterative ray-casting verification across the parameter space.

However, for the specific implementation of HyTouch, we impose a *periodic design constraint* to ensure uniform spatial sampling of the force data. The force-sensing elements are arranged in a regular array with width w and pitch p . Under this constraint, the complex search problem simplifies into a deterministic geometric model. We define the vertical distances from the virtual camera plane to the blocking layer and the target imaging layer as h_b and h_i , respectively, with a geometric magnification factor $\eta = h_i/h_b$.

Consequently, the condition for zero blind spots collapses into two linear inequalities governing the lateral baseline distance D of the virtual cameras:

$$\frac{w}{2} \left(\frac{\eta}{\eta - 1} \right) \leq D \leq \left(p - \frac{w}{2} \right) \left(\frac{\eta}{\eta - 1} \right) \quad (2)$$

The lower bound ensures the virtual camera is wide enough to see past the central obstacle (Center Coverage), while the upper bound ensures it is not so wide as to be blocked by adjacent obstacles (Bilateral Handshake). Substituting our design parameters ($w = 3$ mm, $p = 8$ mm, $\eta \approx 1.30$), (2) yields a feasible range of 6.5 mm $\leq D \leq 28.2$ mm.

b) Simulation-based Verification: Complementing the analytical derivation, an interactive simulation-aided design tool was developed. While the analytical solution provides the theoretical bounds for our periodic design, this tool serves as a

numerical solver based on ray-casting. It allows for the rapid adjustment of parameters (e.g., h_b, h_i, w, p, α) and provides immediate visualization of the light paths and blind spots, as shown in Fig. 6(a).

For the HyTouch prototype, we selected a baseline of $D = 20$ mm, which lies robustly within the analytical feasible region. Fig. 6(b) confirms that this configuration achieves full surface coverage. In contrast, Fig. 6(c) illustrates a failed configuration where suboptimal parameters violate the condition, resulting in shared blind spots. Furthermore, because the tool relies on ray-casting rather than geometric simplification, it essentially provides a general solver capable of evaluating arbitrary layouts, extending its utility beyond the periodic case presented here.

3) Optimized Lighting: To ensure high-quality, high-contrast images, the illumination system is also carefully designed. A key issue in many vision-based tactile sensors is non-uniform lighting and internal reflections. As shown in Fig. 7a(left), a conventional design in which LEDs illuminate the gel through a clear acrylic layer becomes problematic when applied to our multilayer structure. Light rays (yellow arrows) not only produce the desired diffuse reflection from the pigment layer (red arrows) for imaging, but also generate significant noise from diffuse reflection off the underlying force-tactile layer (blue solid arrows) and secondary reflections within the acrylic (blue dashed arrows).

To address this, a custom Light-Guiding Lens is introduced as illustrated in Fig. 7a(right). This new component guides the LED light through total internal reflection, directing it to enter

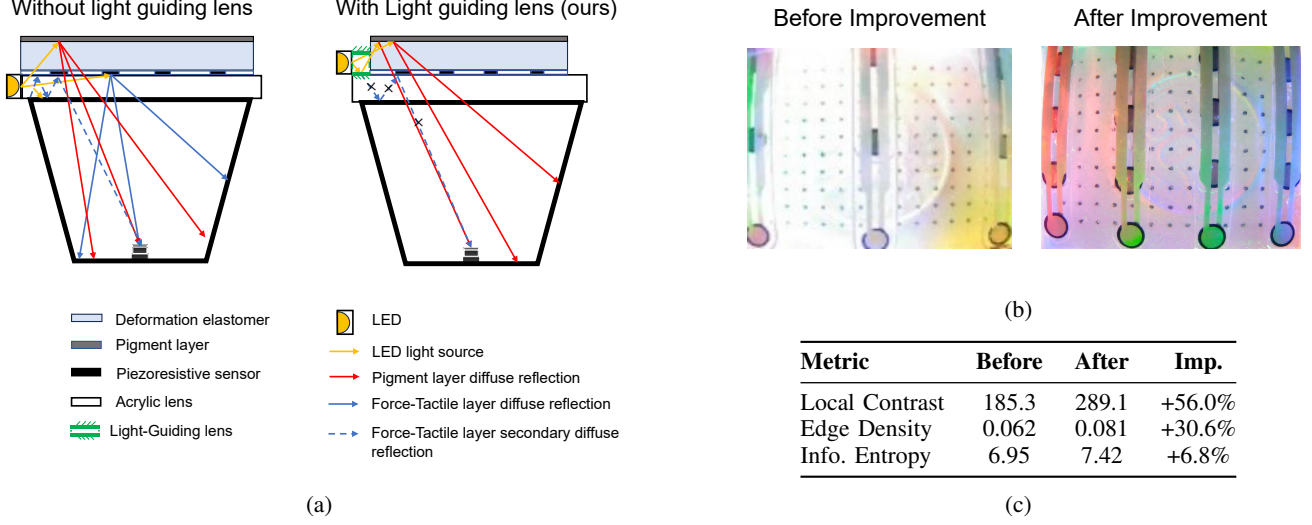


Fig. 7: Optimized lighting system design and verification. (a) Comparison of lighting designs: conventional (left) vs. proposed light-guiding lens (right). (b) Qualitative image quality comparison before (left) and after (right) optimization. (c) Quantitative metrics demonstrating significant improvement in contrast and detail.

the gel from the side. This light path completely avoids direct illumination of the force-tactile layer, thus eliminating the primary source of reflection noise and significantly improving image quality. A light-guiding lens further improves light distribution and also serves as a mechanical fixture that aids in the repeatable alignment of components during reassembly.

As shown in Fig. 7(b), this optimized light path results in a significant visual improvement. Fine textures, such as those on the surface of a coin, become much more clearly discernible. To quantify this enhancement, a computational analysis of the images before and after the optimization is conducted with the results for key metrics summarized in Fig. 7(c). The substantial increases in local contrast and edge density quantitatively confirm the improved ability of the sensor to capture the fine-grained details crucial for dexterous manipulation.

4) *Data Acquisition and Reconstruction Pipeline*: To reconstruct a complete, unoccluded image of the contact surface, a dedicated processing pipeline is developed and visually depicted in Fig. 8.

The process begins with calibrating the fisheye camera to correct for lens distortion using the standard intrinsic matrix and distortion model. The intrinsic camera matrix is given by:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, \quad (3)$$

where f_x and f_y are the focal lengths, and (c_x, c_y) is the optical center. The distortion is then corrected using the following radial and tangential model:

$$x_{\text{distorted}} = x (1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + 2p_1 xy + p_2 (r^2 + 2x^2), \quad (4)$$

$$y_{\text{distorted}} = y (1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + p_1 (r^2 + 2y^2) + 2p_2 xy, \quad (5)$$

where k_1, k_2, k_3 are the radial and p_1, p_2 are the tangential distortion coefficients.

The core logic of the reconstruction is detailed in Algorithm 1. The raw fisheye image is first undistorted. It is then computationally segmented into its three constituent views (left, main, and right). The side views are warped via a homography transformation to align with the main view's perspective.

The final and most critical step is to fill the occluded regions in the main view using data from the warped side views. To avoid visible seams, we employ a gradient-domain fusion technique based on Poisson blending [49]. This is formulated as a minimization problem to find an unknown function f (the final pixel values in the target region Ω) whose gradient ∇f most closely matches the gradient of the source image g within Ω , while matching the boundary $\partial\Omega$:

$$\min_f \iint_{\Omega} |\nabla f - \nabla g|^2 \quad \text{with} \quad f|_{\partial\Omega} = h|_{\partial\Omega}. \quad (6)$$

Solving this standard Poisson problem with Dirichlet boundary conditions ensures that the texture from the source image is seamlessly carried over into the destination, resulting in a visually coherent stitched image.

C. Force-Tactile Subsystem

The force-tactile subsystem is responsible for high-frequency force feedback. Each force-sensing element in the array converts applied normal pressure into a measurable change in electrical resistance. The relationship between the applied pressure P and the resulting resistance R is approximately:

$$R \approx \frac{1}{kP}, \quad (7)$$

where k is a material-dependent sensitivity constant. To linearize this response and convert it to a voltage, a simple ac-

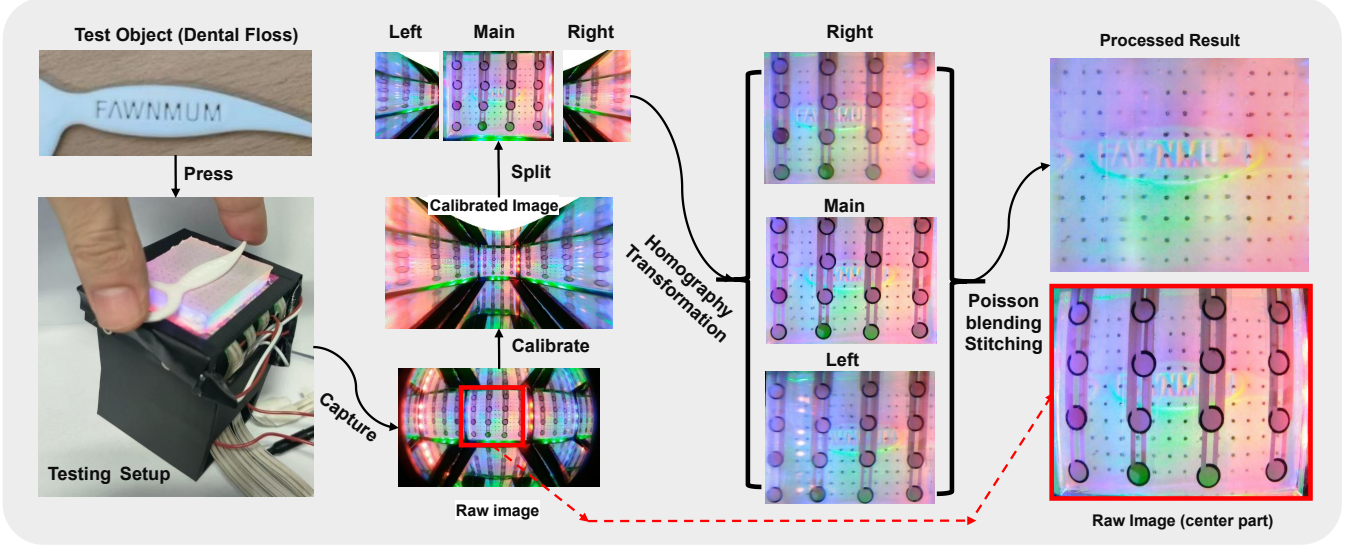


Fig. 8: The in-sensor image processing pipeline for occlusion removal. A raw fisheye image is undistorted, segmented into three views, and then stitched together using the gradient-domain fusion process detailed in Algorithm 1 to form a complete, unoccluded reconstruction.

Algorithm 1 Vision-based Tactile Image Stitching

Require: Raw fisheye image I_{raw} , Camera matrix K , Distortion coeffs D , Homographies $H_{L \rightarrow M}, H_{R \rightarrow M}$, Occlusion masks $Mask_L, Mask_R$

Ensure: Stitched, unoccluded image I_{final}

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 $I_{undistorted} \leftarrow \text{undistort}(I_{raw}, K, D)$ 
 $I_L, I_M, I_R \leftarrow \text{segmentViews}(I_{undistorted})$ 
 $I_{L\_warped} \leftarrow \text{warpPerspective}(I_L, H_{L \rightarrow M})$ 
 $I_{R\_warped} \leftarrow \text{warpPerspective}(I_R, H_{R \rightarrow M})$ 
 $I_{final} \leftarrow I_M$ 
for each occluded patch  $p$  in  $Mask_L$  do
   $center \leftarrow \text{getCenter}(p)$ 
   $I_{final} \leftarrow \text{poissonBlend}(I_{L\_warped}, I_{final}, p, center)$ 
end for
for each occluded patch  $p$  in  $Mask_R$  do
   $center \leftarrow \text{getCenter}(p)$ 
   $I_{final} \leftarrow \text{poissonBlend}(I_{R\_warped}, I_{final}, p, center)$ 
end for
return  $I_{final}$ 

```

quisition circuit is used for each force-sensing element, which implements an inverting operational amplifier. This “virtual ground” circuit produces an output voltage V_{out} that is directly proportional to the force-sensing element’s conductance, and therefore approximately linear with respect to the applied pressure:

$$V_{out} = -V_{ref} \left(\frac{R_f}{R} \right) \propto P. \quad (8)$$

where V_{ref} is the input voltage and R_f is a fixed feedback resistor. This linear relationship allows for straightforward calibration and real-time force measurements.

IV. SENSOR PROTOTYPE AND PERFORMANCE EVALUATION

Following the proposed sensor design detailed in the previous section, a HyTouch sensor prototype has been built as shown in Fig. 1.

Significantly, a 4×4 layout is adopted for the force-sensing array design in this prototype. In the reflective mirror system design, the feasible theoretical range for the virtual baseline is calculated as $6.5 \text{ mm} \leq D \leq 28.2 \text{ mm}$ based on the design inequality (2) with the sensor parameters ($w = 3 \text{ mm}$, $p = 8 \text{ mm}$, and $\eta \approx 1.30$). Furthermore, to maximize robustness against assembly tolerances, a baseline of $D = 20 \text{ mm}$ is selected for this prototype, which lies near the center of this feasible region. Consequently, the planar mirrors are installed at an inclination angle $\alpha = \arctan(D/Z_{off}) \approx 76^\circ$, where Z_{off} is the vertical offset of the virtual cameras. Finally, the full coverage of the tactile surface for the final design ($D = 20 \text{ mm}$) is confirmed by the developed simulation-based design tool with the result shown in Fig. 6(a).

In the following of this section, a series of experiments is designed to quantify the fundamental performance metrics of the HyTouch sensor. Here, the sensor’s force response and linearity are characterized, the sensor’s ability to detect high-frequency vibrations is evaluated, and the accuracy of the reconstruction pipeline described in Section III-B is assessed.

A. Force Sensing Calibration and Linearity

To characterize the force-tactile layer’s response, a systematic force calibration experiment is conducted.

1) *Experimental Setup:* Figure 9 depicts the experimental setup, where a KUKA iiwa robotic arm equipped with an ATI F/T sensor Mini45 and a cylindrical probe is used. This

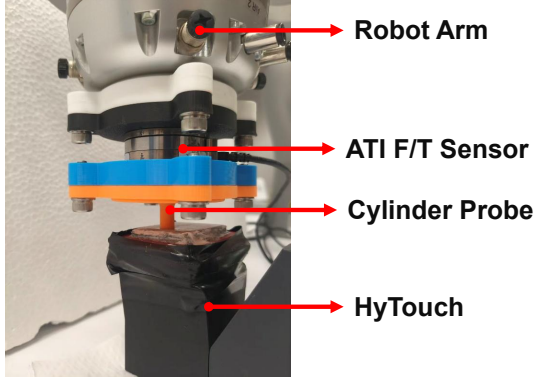


Fig. 9: Experimental setup for the force sensing calibration.

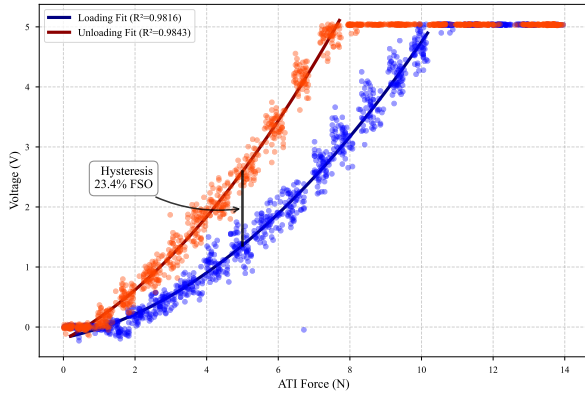


Fig. 10: Force calibration curve of a single taxel, averaged over three loading-unloading cycles.

setup allows for the application of precise and repeatable force profiles to the developed tactile sensor.

2) *Procedure*: The probe is programmed to execute three consecutive loading-and-unloading cycles on a single taxel. In each cycle, the robot presses down for a duration of 5 seconds, holds the position for 1 second, retracts for another 5 seconds, and pauses for 1 second before the next cycle. Throughout this process, the data from both the ATI sensor (ground truth force) and the HyTouch's taxel (output voltage) were recorded synchronously.

3) *Results and Analysis*: The experimental results with the average over three cycles are presented in Fig. 10, where the plot shows the raw data points, the fitted quadratic models for both loading and unloading phases, and the calculated hysteresis. Significantly, to accurately model this behavior, quadratic polynomials are used to fit the data from the loading and unloading phases separately. As can be seen, the HyTouch exhibits a distinct and highly repeatable response.

Moreover, the analysis reveals that the quadratic model provides an excellent fit for both phases, confirmed by high R-squared values. For the loading phase, the relationship is described by

$$V = 0.0370F^2 + 0.1228F - 0.1791 \quad (9)$$

with an R^2 value of 0.9816. The unloading phase is modeled by

$$V = 0.0479F^2 + 0.3215F - 0.2153 \quad (10)$$

with an R^2 value of 0.9843. However, the response exhibits clear nonlinearity, primarily due to the intrinsic behavior of visual-tactile layer. This nonlinearity results in hysteresis, a key characteristic of viscoelastic sensors, was quantified at 5.0 N as a voltage difference of 1.23 V, corresponding to 23.37% of the Full-Scale Output (FSO).

While a hysteresis of 23.37% is significant, it is a common trait of soft elastomer-covered sensors. Crucially, this does not hinder the sensor's utility in our target applications. For the high-speed cable following task, the control strategy utilizes the taxels primarily as logic-based triggers (active/inactive) to determine contact location, rendering absolute force precision less critical. For the material classification task, the deep learning model inherently learns to compensate for these nonlinearities, including hysteresis, during the training phase. This comprehensive characterization provides a reliable model for describing the sensor's force-voltage behavior and supports accurate interpretation of force measurements in real-world applications.

B. High-Frequency Vibration Response

One hypothesis of the proposed sensor design is that the elastomer layer can provide visual compliance without excessively damping the high-frequency signals intended for the force-tactile array.

1) *Experimental Setup*: In this experiment, the setup is shown in Fig. 11(a). As can be seen, a Faulhaber LM1483 linear motor is fixed vertically while its linear mover is directly in contact with the HyTouch's surface.

2) *Procedure*: To validate this, the sensor's response to vibrations is tested with vibrations generated by the linear motor operating at 25 Hz and 50 Hz (see Fig. 11(a)). To accurately quantify the signal attenuation, a ground-truth force-sensing element was placed on the outer surface of the gel, directly under the motor probe, while the sensor's internal force-sensing element measured the transmitted force.

3) *Results*: The experimental results are presented in Fig. 11(b)–(e). It is clear that the sensor's internal force-sensing element can successfully measure the vibrations at both frequencies. By comparing the signal amplitudes, the attenuation is found to be only 9.71% at 25 Hz and 11.64% at 50 Hz. These experimental results quantitatively confirm that the 8-mm-thick elastomer allows for the effective transmission of high-frequency force data, validating a core principle of the proposed integration design.

C. Comparison with Other Sensors

Finally, Table I presents a comparative analysis of HyTouch against several state-of-the-art tactile sensors. The results show that HyTouch is distinctive in providing both high-density vision-based tactile feedback (560×450 resolution) and high-frequency force response (up to 300 Hz). This combination, supported by the quantitative characterization results presented

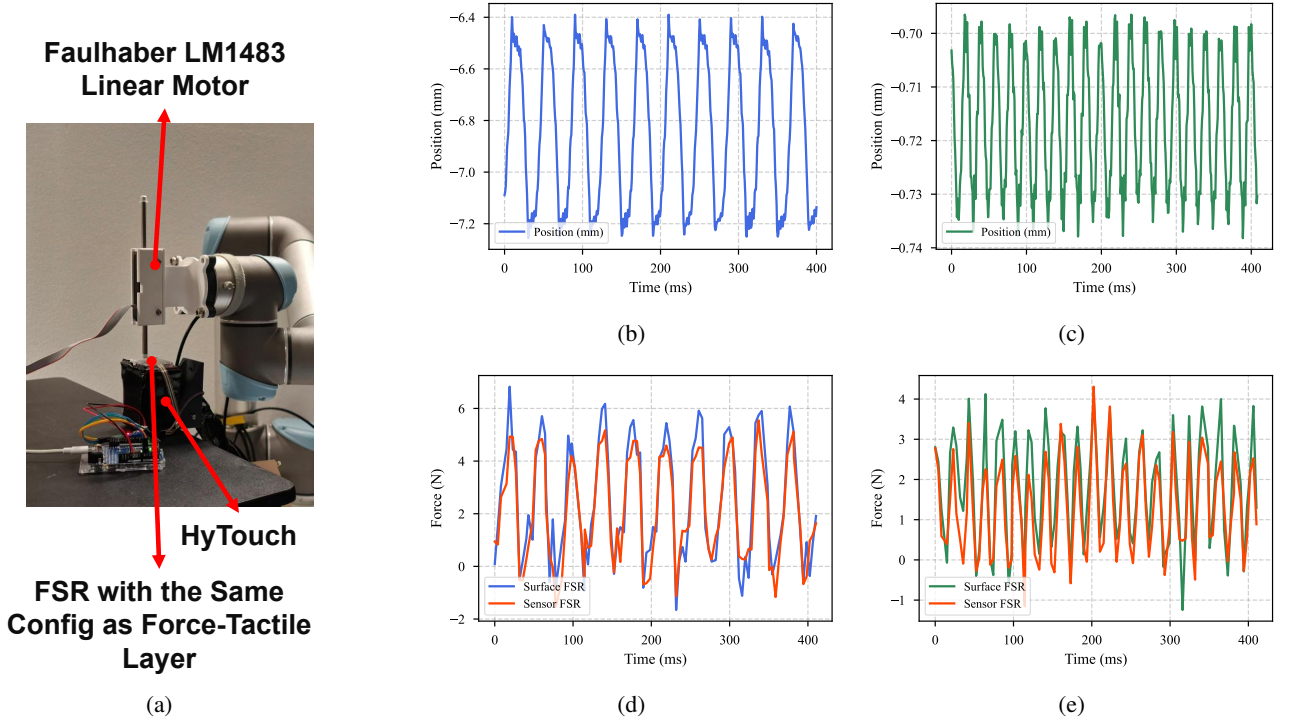


Fig. 11: Validation of high-frequency response. (a) Experimental setup with a linear motor and a surface force-sensing element for ground-truth measurement. (b) Motor displacement at 25 Hz. (c) Motor displacement at 50 Hz. (d) Sensor response at 25 Hz (surface force-sensing element vs. internal force-sensing element). (e) Sensor response at 50 Hz (surface force-sensing element vs. internal force-sensing element).

TABLE I: Comparison of HyTouch with Other Tactile Sensors

Sensor	Modalities	Max Freq.	Density	Range	High-Res Surface	High-Freq Dynamic	Distributed Force
GelSight [4]	Vision (Shading)	60 Hz	640×480	NA	✓	✗	✗
TacTip [5]	Vision (Marker)	90 Hz	~127 Markers	NA	✓	✗	✗
EveTac [46]	Event Vision	>1 kHz	640×480	NA	✓	✓	✗
DIGIT 360 [44]	Vision + Force	1 kHz	240×320	0–10 N	✓	✓	✗
BioTac [42]	Imp/Therm/Vib	1040 Hz	19 El.	0–50 N	✗	✓	✗
XELA [?]	Force Array	100 Hz	4×4	0–20 N	✗	✓	✓
GTac [9]	Mag. + Array	250 Hz	4×4	0–2 N	✗	✓	✓
HyTouch	Vision + Array	300 Hz	560×450	0–8 N	✓	✓	✓

in this section, enables a wide range of applications that are difficult to achieve with single-modality sensors or less integrated multimodal designs.

V. APPLICATION IN ROBOTIC MANIPULATION TASKS

After evaluating the sensor’s fundamental capabilities in the previous section, this section demonstrates how these capabilities translate into superior performance in complex robotic manipulation. Two representative tasks are chosen: the high-speed control task and the perceptual classification task. Both are designed to show that the co-location of High-Resolution Surface Sensing, High-Frequency Dynamic Response, and Spatially Distributed Force Sensing enables novel control and fusion strategies that are impossible for sensors that compromise on any vertex of the trilemma.

A. Task 1: High-Speed Cable Following

1) *Task Objective:* Cable manipulation is a typical robotics task that requires precise shape estimation and localization as well as rapid dynamic response. Previous work using only a vision-based tactile sensor, GelSight, was limited to a following speed of 6.5 cm/s due to the image processing latency [1]. Here, a hybrid control strategy is developed by leveraging HyTouch’s multimodal data to significantly surpass this speed limit by overcoming the inherent speed-accuracy trade-off of single-modality sensing.

2) *Experimental Setup:* The experimental setup for this cable following task is shown in Fig. 12, which consists of a UR5 robotic arm with a Robotiq 2F-85 gripper, on which the HyTouch sensor is mounted. The control system runs on a computer (Intel Core i9 CPU, NVIDIA RTX 4060 GPU), which communicates with the robotic system and the sensor

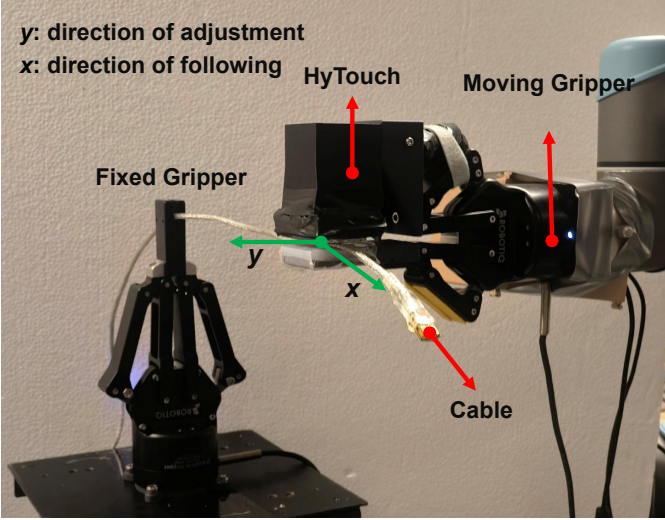


Fig. 12: Experimental setup for the high-speed cable following task.

via ROS. The robot motion is commanded using the `servoL` function with a lookahead time of 0.1 s and a gain of 300 for smooth tracking.

3) *Control Strategies*: In this task, the robot starts in the *following* phase, where it tracks the cable contour online using tactile feedback. During following, we continuously monitor the lateral offset error y of the cable relative to the gripper centerline (Fig. 14). As $|y|$ grows, the cable-sensor contact can enter a strongly nonlinear regime (e.g., partial loss of contact, rolling/sliding, and changing contact patch), violating the assumptions of the contact model used for control and potentially causing the controller to fail. To prevent such failure and avoid cable slippage, we introduce an *adjustment* phase as a recovery mechanism. When $|y|$ exceeds a threshold of 1.2 cm, the robot stops the following motion, moves the gripper back to a predefined home pose, and applies a lateral correction $\Delta y = -y$ to compensate the accumulated error. After this recovery motion, the robot resumes the following phase.

The following control strategy and the adjustment strategy for each single modality (visual-tactile vs. force-tactile) are shown below. The visual-tactile modality computes y, θ , and α , from the reconstructed contact geometry (Fig. 14, top), while the force-tactile modality estimates y from a weighted aggregation of the 4×4 force map (Fig. 14, bottom), which is faster but less precise.

- **Visual-Tactile Modality with Linear Quadratic Regulator (LQR)**: This strategy re-implements the state-of-the-art method from [1], serves as the baseline. It only uses the high-resolution visual-tactile stream (see Fig. 14, right-top) for precise state estimation of the cable's lateral offset (y), orientation (θ), and curvature (α), as shown in Fig. 13. A linear model of the system dynamics, as shown below, was identified, and an LQR was designed to compute the optimal control input based on the system

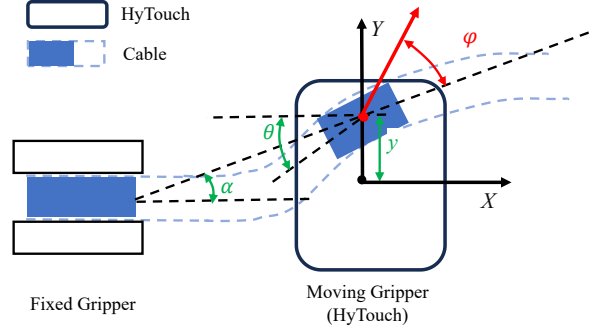


Fig. 13: The cable manipulation dynamics model, adapted from [1]. The state of the cable relative to the sensor is described by its lateral offset (y), its orientation (θ), and its local curvature (α).

model and visual-tactile sensing information.

$$\dot{x} = Ax + Bu, \quad (11)$$

where x represents the contact states, A is the state matrix, B is the input matrix, and u is the system input. However, the control frequency is limited to 12 Hz because of the computational effort required during image processing. For adjustment, the controller uses the visual-tactile estimate of y and triggers recovery when $|y| > 1.2$ cm, executing the same $\Delta y = -y$ correction after returning to the home pose.

- **Force-Tactile Modality with PID Controller**: This strategy relies on the high-frequency force-tactile stream (see Fig. 14, bottom). The raw 4×4 pressure map is thresholded and discretized to compute a 1D estimation of the cable's lateral position, which serves as an error signal for a PID controller governed by

$$v_y(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}, \quad (12)$$

where K_p , K_i , and K_d are the PID controller parameters, e is the error signal corresponding to the estimated cable position y , and v_y is the PID controller output. For adjustment, the same $|y| > 1.2$ cm trigger and $\Delta y = -y$ recovery are used, but with y estimated from the force-tactile force map. Significantly, the high feedback-loop sampling frequency of 300 Hz enables rapid response, but the spatial resolution is low.

- **Multimodality with Hybrid Controller (Ours)**: The proposed controller uses the force-tactile PID strategy during the following phase to maximize tracking speed. Once $|y| > 1.2$ cm, it switches to the adjustment phase and performs recovery using the more accurate visual-tactile estimate of y (i.e., the same $\Delta y = -y$ correction, but with visual-tactile state estimation). This phase switching combines the high-speed reactivity of the array with the high-precision recovery of vision-based tactile sensing.

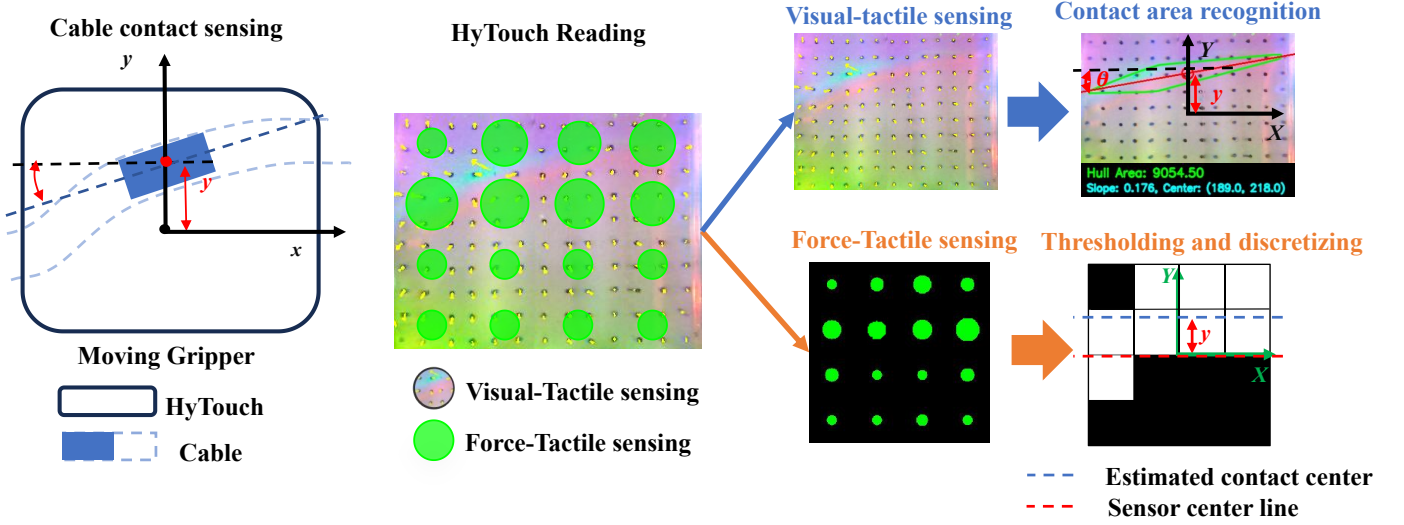


Fig. 14: Sensing and processing pipelines for the cable following task. The sensor’s raw multimodal reading is processed in two parallel streams. The visual-tactile stream (top) provides high-resolution contact area information for precise state estimation. The force-tactile stream (bottom) provides a high-frequency, low-resolution estimate of the cable’s center line for reactive control.

TABLE II: Quantitative Performance in High-Speed Cable Following

Target Velocity (cm/s)	Modality	Avg. Speed (cm/s)	Adjustment Counts	Total Time (s)
3	Visual-tactile	2.42	4.0	80.31
	Force-Tactile	2.70	5.8	85.71
	Hybrid (Ours)	2.73	2.4	60.14
6	Visual-tactile	5.25	5.2	58.64
	Force-Tactile	5.29	5.5	65.98
	Hybrid (Ours)	5.49	2.6	47.90
18	Visual-tactile	NA	NA	NA
	Force-Tactile	15.0	3.6	36.60
	Hybrid (Ours)	15.0	1.6	27.70
24	Visual-tactile	NA	NA	NA
	Force-Tactile	19.7	2.1	22.83
	Hybrid (Ours)	20.4	1.3	19.59

4) *Results:* All three control strategies are evaluated at four different target velocities (3, 6, 18, and 24 cm/s) over a 1-meter cable using the same experimental setup, and the experimental results are summarized in Table II.

a) *Performance at Low Speeds (3 and 6 cm/s):* At low speeds, all three controllers can complete the task successfully. However, the Hybrid controller consistently requires the fewest corrective adjustments, resulting in the shortest overall task completion times. In contrast, the force-tactile-only controller, which relies on a coarse map for recovery, triggers the largest number of adjustments.

b) *Performance at High Speeds (18 and 24 cm/s):* At high speeds, the performance differences became significant. The visual-tactile modality results in task failure (denoted as NA in Table II), because its low control frequency of approximately 12 Hz is insufficient to capture rapid changes in the cable’s state, leading to instability and cable slippage from the gripper. In contrast, both the Force-Tactile and Hybrid controllers complete the task successfully, with the Hybrid approach performing the best. In other words, the proposed

method can achieve three times faster than the state-of-the-art method [1].

At the maximum speed of 24 cm/s, the proposed hybrid controller completes the task in 19.59 s with only 1.3 adjustments on average, compared with 22.83 s and 2.1 adjustments for the force-tactile controller. This corresponds to a 14.2% reduction in time and a 38.1% reduction in adjustments, reflecting the benefits of combining high-frequency tracking with high-accuracy correction enabled by the proposed sensor and control strategy. In addition, due to the fast following speed and the efficient adjustment mechanism, the proposed method achieves average speeds that are closest to the target velocities.

B. Task 2: Tool-Mediated Material Classification

1) *Task Objective:* Inferring material properties through a rigid tool is a useful but challenging robotic task that requires precise extrinsic contact sensing. Prior approaches that rely on a single sensing modality often struggle because quasi-static deformation cues or high-frequency contact dynamics

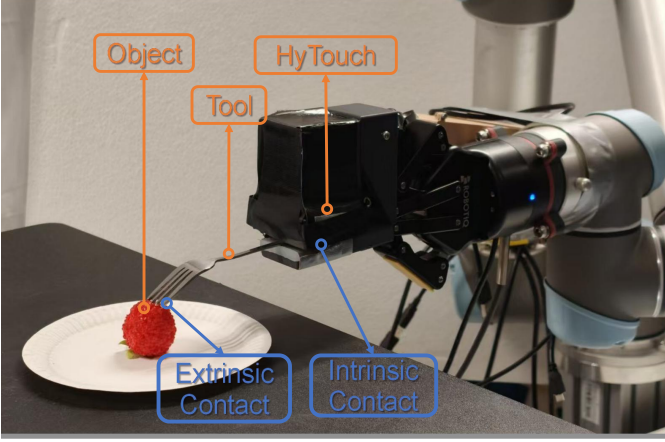


Fig. 15: Experimental setup for the tool-mediated material classification task. The diagram illustrates the concepts of intrinsic (tool-sensor) and extrinsic (tool-object) contact.

alone may be insufficient for reliable discrimination. Here, a multimodal strategy is adopted by leveraging HyTouch’s combined vision-based deformation information and high-frequency force response to overcome these limitations. The fused representation is expected to improve classification accuracy by resolving ambiguities that arise when only one modality is used.

2) *Experimental Setup*: The experimental setup for this task is shown in Fig. 15. A 2-finger gripper holds one end of a tool (e.g., fork or spoon) to establish intrinsic contact with the HyTouch sensor, while the tool tip at the other end makes extrinsic contact with various objects. The specific objects and tools used in this experiment are shown in Fig. 16.

3) *Data Collection*: To train the material classification model, a dedicated dataset was collected. A dataset of 1,600 samples was constructed by performing 400 pressing trials per object (200 using the fork and 200 using the spoon). Each trial executes a 4-second motion over a fixed 8 mm distance with the velocity profile randomly selected from linear, quadratic, exponential, or sinusoidal trajectories to ensure diversity. During each press, the sensor captured two synchronized data streams: a 4×4 300 Hz force array sequence (collecting the last 600 sensing step) by the force-tactile modality and a set of two high-resolution tactile images by the visual-tactile modality before contact and after the contact is stable.

4) *Classification Models and Training*: By leveraging different sensing modalities, three deep learning-based material classification models are developed, with their architectures illustrated in Fig. 17. The details of each model are as follows:

- **Visual-Tactile Model**: The difference between the first and last tactile images (224×224 pixels) is computed to represent the overall deformation. The deformation representation is then passed through a frozen, pre-trained ResNet-18 to extract a 512-dimensional feature vector, followed by a final fully connected layer classifier.



Fig. 16: The four objects (hard silicone, soft silicone, sponge, soft doll) and the two tools (fork, spoon) used in the tool-mediated material classification task.

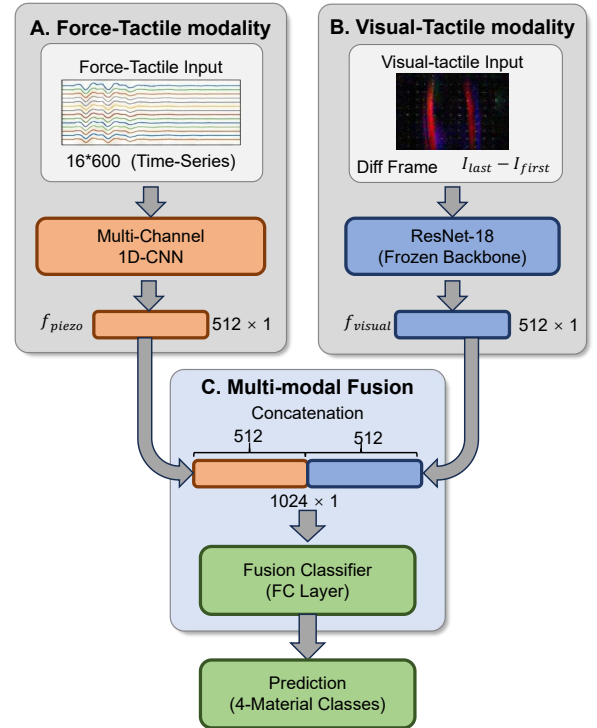


Fig. 17: The three classification model architectures. (a) The Visual-tactile model uses a ResNet-18 backbone on difference images to capture overall deformation. (b) The Force-Tactile model uses a multi-channel 1D-CNN on time-series data to learn dynamic signatures. (c) Our proposed Hybrid Fusion model concatenates the feature embeddings from both modalities to form a richer representation for the final classifier.

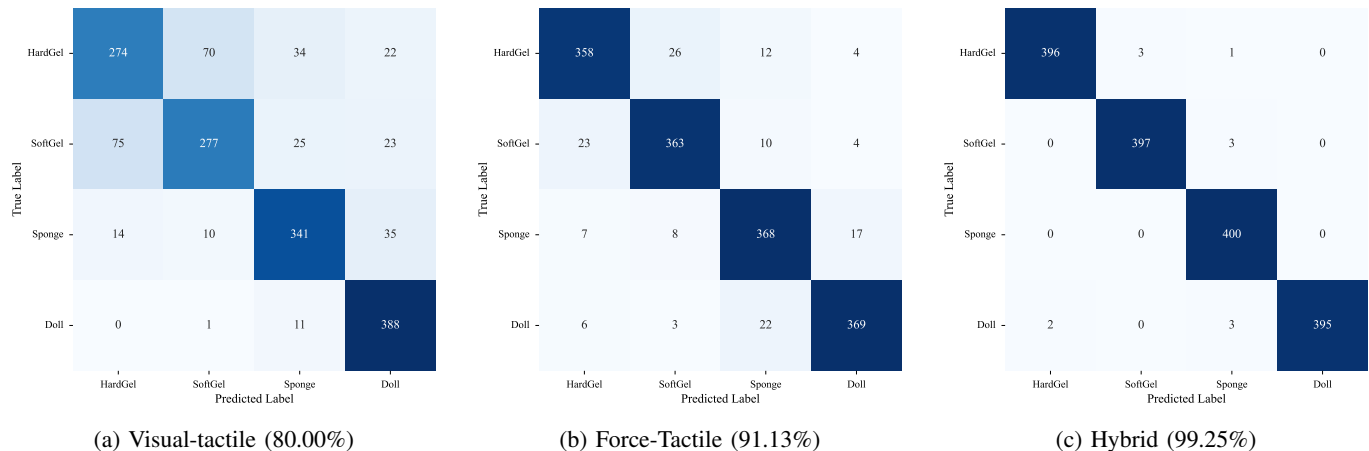


Fig. 18: Aggregated confusion matrices over 8 folds for the three classification models. The near-perfect diagonal structure of the Hybrid model’s matrix (c) shows its ability to resolve the ambiguities present in the single-modality classifiers (a, b).

- **Force-Tactile Model:** The 16-channel, 600-timestep force sequence is processed using a multi-channel 1D convolutional neural network (1D-CNN) designed to learn the temporal dynamics. This network also outputs a 512-dimensional feature vector for a final fully collected layer classifier.
- **Hybrid Model:** The 512-dimensional feature vectors from both modalities (obtained using the aforementioned models) are concatenated into a unified 1024-dimensional representation, which is then fed into a final fully connected layer for classification.

5) *Results:* The classification accuracies of all the models are evaluated using 8-fold stratified cross-validation. The performance of each model is summarized in Table III, and the aggregated confusion matrices are shown in Fig. 18. As can be seen, the Hybrid model achieved a near-perfect accuracy of 99.25%, significantly outperforming both the Force-Tactile (91.13%) and Visual-Tactile (80.00%) models.

Furthermore, the confusion matrices highlight the complementary strengths of the two modalities. The Visual-Tactile model has difficulty distinguishing between the two silicone materials, frequently misclassifying HardGel as SoftGel (70 instances) and SoftGel as HardGel (75 instances). This indicates that their similar stiffness produces nearly indistinguishable deformation patterns. The Force-Tactile model performs better but still exhibits confusion between the gels and other types of materials, suggesting that dynamic force characteristics alone are not sufficiently distinctive for all classes.

In contrast, the Hybrid model achieves a near-perfectly diagonal confusion matrix, demonstrating that the multimodal fusion resolves the ambiguities inherent in each individual modality. The force signals help differentiate “visually” similar silicones (with similar deformations), while the deformation images separate materials with overlapping dynamic responses. This result provides strong evidence that integrating both modalities leads to a more comprehensive and accurate material understanding.

TABLE III: Tool-Assisted Material Classification Accuracy

Modality	Classification Accuracy (%)
Visual-tactile (ResNet)	80.00 ± 2.2
Force-Tactile (1D-CNN)	91.13 ± 1.4
Hybrid (Ours)	99.25 ± 1.0

VI. CONCLUSION

In this paper, a new hardware integration concept for tactile sensing is introduced. The design addresses the long-standing “tactile trilemma” by co-locating and decoupling multiple sensing modalities to simultaneously achieve high-resolution spatial tactile imaging, high-frequency dynamic response, and spatially distributed force measurement. Building on this principle, a bio-inspired multimodal tactile sensor, HyTouch, is developed, featuring a compact optical system engineered to minimize inter-modality interference. Furthermore, the prototype of the HyTouch is built, and comprehensive characterization experiments are conducted to quantify the sensor’s performance. Moreover, two challenging robotic tasks are conducted using HyTouch, alongside comparisons with state-of-the-art methods and single-modality baselines. These evaluations clearly demonstrate the advantages of the integrated sensing approach. The experimental results provide strong support for the central hypothesis. In high-speed cable following, the Hybrid controller leverages HyTouch’s ability to provide high-frequency force-distribution signals for rapid tracking, capabilities unattainable with single-channel force sensors, while relying on high-resolution vision for fine-grained adjustments. In the tool-mediated material classification task, the model integrating both modalities achieves near-perfect accuracy (99.25%), resolving perceptual ambiguities that confound single-modality classifiers. These findings confirm that the proposed hardware integration method enables a more complete and reliable perceptual understanding than either sensing modality alone.

This work establishes the foundation for a new class of adaptive, high-performance tactile sensors. As illustrated in Fig. 19, the proposed design concept is flexible and can be

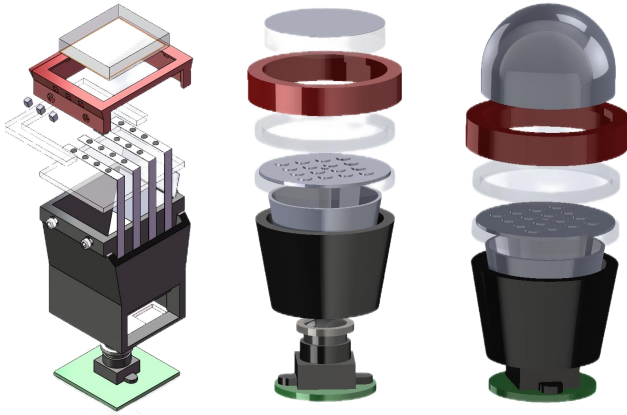


Fig. 19: The proposed integration method allows for flexible design configurations. The density and arrangement of the force-sensing array, as well as the shape and material of the visual-tactile layer, can be adapted to specific task requirements.

tailored to a broad range of robotic manipulation challenges. Although this paper presents one specific prototype, the underlying integration strategy is generalizable. Future designs may vary in force-array density, sensing-layer materials, or geometric configurations to meet specialized task requirements.

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